

ABSTRACT

A Educational reform in the digital learning age increasingly calls for systems not only to track but also to amplify student participation in blended-classroom settings. Toward United Nations Sustainable Development Goal (SDG) 4: Quality Education, this work presents PrivLogEdu, a novel method for real-time engagement detection from non-visual log data with high prediction accuracy and strong privacy protection. Differing from vision-based emotion detection methods that pose privacy and deployment issues, PrivLogEdu leverages behavioral and temporal learning logs problem-solving sequences, time-on task, and skill-switching patterns to infer levels of engagement without recording personally identifiable visual information. The suggested approach makes use of a multi-stage pipeline: data preprocessing, sophisticated feature engineering, Skill2Vec based embedding extraction, and a GRU-attention deep neural architecture tailored for sequence modeling. This architecture allows the system to learn temporal dependencies, pay attention to important behavioral features, and provide engagement classification at three levels (low, moderate, high) with an accuracy of 99.91% on the EdNet-KT1 dataset. Additionally, the model uses class balancing, label smoothing, and threshold tuning to ensure strong generalization and low misclassification. Experiment assessment proves to be more effective compared to traditional engagement recognition models, with near-optimal F1-scores and slight computational costs that are acceptable for real-time utilization. Comparative analysis also attests to the fact that PrivLogEdu outperforms baseline LSTM and Transformer models in accuracy and stability. In addition, avoiding visual modalities, the system naturally minimizes ethical, and data protection risks and fits with modern educational policies prioritizing learner privacy. This research not only helps advance AI-based adaptive learning systems but also provides a scalable solution for learning platforms looking to enhance learning outcomes in hybrid and remote environments while maintaining privacy standards.